

90-DAY DSA MASTER ROADMAP

A Rigorous, Python-First Curriculum Engineered for MNC & Product-Based Backend Software Engineering Interviews

Target Role: Python Backend / Software Engineer

Start Date: September 9, 2026

Weekday Time Commitment: 2.5 - 3.0 Hours/Day

Target Problem Count: ~230 Selected Problems

Total Duration: 90 Days (12.8 Weeks)

End Date: December 7, 2026

Weekend Time Commitment: 5.0 Hours/Day

Primary Language: Python 3.11+

Curriculum Architecture Highlights

- **Zero Hand-Waving:** Complete coverage of all 90 days with daily scheduled focus areas and problem targets.
- **Python-First Mastery:** Built entirely around idiomatic Python (built-ins, standard library modules like collections, heapq, bisect, and recursion memory profiling).
- **Interview-Centric Strategy:** Focused exclusively on pattern recognition, time-bounded problem-solving, and clean verbal articulation.
- **Spaced Repetition & Re-Solving:** Strict revision loops (3-day, 7-day, 21-day) to build intuition rather than rote memory.
- **Realistic Time Budgeting:** Precision workloads structured explicitly for working professionals (2.5-3 hrs M-F, 5 hrs Sat-Sun).

1. Executive Summary & How to Use This Roadmap

This curriculum is engineered for backend developers aiming to clear coding interviews at Tier-1 MNCs, high-growth product companies, and top-tier startups. It treats Data Structures and Algorithms not as abstract competitive programming puzzles, but as practical engineering patterns.

The Daily Execution Loop: On weekdays, allocate 20 minutes to theory/pattern recognition, 60-90 minutes to active coding, 20 minutes to debugging/dry-running, and 15 minutes to documenting trade-offs in your mistake log. On weekends, use the 5-hour allocation for deep-dive topics, timed weekly assessments, and spaced-repetition re-solving.

2. 90-Day Architecture & Phase Breakdown

Phase	Dates / Days	Focus & Objectives	Target Problems
Phase 1: Foundations	Sept 09 - Sept 22 (Days 1-14)	Python DSA primitives, complexity, Arrays, Strings, Two Pointers, Sliding Window, Prefix Sum.	35 Easy/Medium
Phase 2: Core Linear & Non-Linear	Sept 23 - Oct 13 (Days 15-35)	Linked Lists, Stacks, Queues, Monotonic Structures, Binary Search, Trees, BSTs, Heaps/Top-K.	55 Medium/Hard
Phase 3: Graphs & Recursion	Oct 14 - Nov 03 (Days 36-56)	Backtracking, Graph Traversals (BFS/DFS), Topological Sort, Union-Find, Shortest Paths (Dijkstra), Tries.	50 Medium/Hard
Phase 4: Dynamic Programming & Greedy	Nov 04 - Nov 24 (Days 57-77)	Greedy Intervals, 1D DP, 2D Grid DP, Knapsack, Subsequences, State Machine DP.	50 Medium/Hard
Phase 5: Interview Bootcamp & Final Prep	Nov 25 - Dec 07 (Days 78-90)	Mixed problem sets, timed mock interviews, weak spot repair, speed optimization, final assessment.	40 Mixed Medium/Hard

3. Python-Specific DSA Essentials

Module / Feature	Key Use Cases in DSA	Time Complexity Highlights
collections.deque	Double-ended queue. Essential for BFS and Sliding Window Monotonic Queues.	Append/Pop left/right: O(1) Index lookup: O(N)
heapq	Min-Heap by default. Invert values for Max-Heap. Essential for Top-K and Dijkstra.	heappush / heappop: O(log N) heapify: O(N)
bisect	Binary search on sorted lists. bisect_left, bisect_right.	Search: O(log N) Insertion: O(N) due to shift
functools.lru_cache	Automatic memoization for Top-Down Dynamic Programming.	O(1) overhead per lookup
collections.defaultdict	Graph adjacency lists, grouping anagrams, hash maps with default collections.	Key lookup/insert: O(1) avg

Idiomatic Python Template: Binary Search on Answer

```
def binary_search_on_answer(arr, target_condition):  
    low, high = min_possible_val, max_possible_val  
    ans = low  
    while low <= high:  
        mid = (low + high) // 2  
        if is_feasible(mid, arr, target_condition):  
            ans = mid # Feasible! Save answer and try to optimize further  
            high = mid - 1  
        else:  
            low = mid + 1 # Infeasible, increase bound  
    return ans
```

4. Complete 90-Day Daily Roadmap

DAY 1: Sep 09, 2026 (Wed)	Phase 1: Foundations
Topic Focus:	Big-O Analysis & Python Internals
Theory / Pattern:	Time/Space Complexity, Recursion Limit, List vs Deque Memory
Target Problems:	Two Sum (LeetCode #1), Valid Anagram (#242)
Allocated Time:	2.5 hrs
DAY 2: Sep 10, 2026 (Thu)	Phase 1: Foundations
Topic Focus:	Arrays & Two Pointers - Opposite Direction
Theory / Pattern:	Left/Right pointer convergence on sorted arrays
Target Problems:	Two Sum II (#167), 3Sum (#15), Container With Most Water (#11)
Allocated Time:	2.5 hrs
DAY 3: Sep 11, 2026 (Fri)	Phase 1: Foundations
Topic Focus:	Two Pointers - Fast & Slow Pointer
Theory / Pattern:	Cycle detection, In-place array modification
Target Problems:	Remove Duplicates (#26), Move Zeroes (#283), Happy Number (#202)
Allocated Time:	2.5 hrs
DAY 4: Sep 12, 2026 (Sat)	Phase 1: Foundations
Topic Focus:	Sliding Window - Fixed Size
Theory / Pattern:	Maintain window state across contiguous subarrays
Target Problems:	Maximum Average Subarray I (#643), Permutation in String (#567)
Allocated Time:	2.5 hrs
DAY 5: Sep 13, 2026 (Sun)	Phase 1: Foundations
Topic Focus:	Sliding Window - Variable Size (Max/Min)
Theory / Pattern:	Expand right to satisfy condition, shrink left to optimize
Target Problems:	Minimum Size Subarray Sum (#209), Longest Substring Without Repeating Characters (#3)
Allocated Time:	2.5 hrs
DAY 6: Sep 14, 2026 (Mon)	Phase 1: Foundations
Topic Focus:	Weekend Mastery: Two Pointers & Sliding Window
Theory / Pattern:	Deep dive on edge cases, shrink conditions, multi-character sets
Target Problems:	Longest Repeating Character Replacement (#424), Subarrays with K Different Integers (#992)
Allocated Time:	5.0 hrs

DAY 7: Sep 15, 2026 (Tue)	Phase 1: Foundations
Topic Focus:	Weekend Revision & Timed Test 1
Theory / Pattern:	Weekly Review 1 + 60-min Timed Assessment on Two Pointers & Windows
Target Problems:	Re-solve missed problems; Test Set: 1 Easy, 2 Mediums
Allocated Time:	5.0 hrs
DAY 8: Sep 16, 2026 (Wed)	Phase 1: Foundations
Topic Focus:	Prefix Sum Fundamentals
Theory / Pattern:	Range sum queries in $O(1)$ after $O(N)$ precomputation
Target Problems:	Range Sum Query - Immutable (#303), Subarray Sum Equals K (#560)
Allocated Time:	2.5 hrs
DAY 9: Sep 17, 2026 (Thu)	Phase 1: Foundations
Topic Focus:	Prefix Sum + Hash Map Combinations
Theory / Pattern:	Handling negative numbers and modulo prefix sums
Target Problems:	Continuous Subarray Sum (#523), Subarray Sums Divisible by K (#974)
Allocated Time:	2.5 hrs
DAY 10: Sep 18, 2026 (Fri)	Phase 1: Foundations
Topic Focus:	Strings & Hashing Techniques
Theory / Pattern:	Custom hash keys, frequency maps, character array manipulation
Target Problems:	Group Anagrams (#49), Minimum Window Substring (#76)
Allocated Time:	2.5 hrs
DAY 11: Sep 19, 2026 (Sat)	Phase 1: Foundations
Topic Focus:	Difference Array Pattern
Theory / Pattern:	Range updates in $O(1)$ time across array ranges
Target Problems:	Corporate Flight Bookings (#1109), Car Pooling (#1094)
Allocated Time:	2.5 hrs
DAY 12: Sep 20, 2026 (Sun)	Phase 1: Foundations
Topic Focus:	Matrix Manipulation & Traversals
Theory / Pattern:	2D grid indexing, in-place rotations, spiral orders
Target Problems:	Spiral Matrix (#54), Rotate Image (#48), Set Matrix Zeroes (#73)
Allocated Time:	2.5 hrs

DAY 13: Sep 21, 2026 (Mon)	Phase 1: Foundations
Topic Focus:	Weekend Focus: Advanced Strings & Matrix
Theory / Pattern:	String matching algorithms (KMP/Rabin-Karp concepts) & Grid transformations
Target Problems:	Find All Anagrams in a String (#438), Word Search Grid Prep (#79)
Allocated Time:	5.0 hrs
DAY 14: Sep 22, 2026 (Tue)	Phase 1: Foundations
Topic Focus:	Weekend Revision & Timed Test 2
Theory / Pattern:	Weekly Review 2 + 60-min Timed Assessment on Prefix Sums & Matrices
Target Problems:	Re-solve missed problems; Test Set: 3 Mediums
Allocated Time:	5.0 hrs
DAY 15: Sep 23, 2026 (Wed)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Linked List Core & Two Pointers
Theory / Pattern:	Singly/Doubly linked lists, dummy nodes, pointer mutation
Target Problems:	Reverse Linked List (#206), Merge Two Sorted Lists (#21)
Allocated Time:	2.5 hrs
DAY 16: Sep 24, 2026 (Thu)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Linked List Cycle & Fast/Slow Pointers
Theory / Pattern:	Floyd's Cycle Detection algorithm and intersection discovery
Target Problems:	Linked List Cycle II (#142), Reorder List (#143)
Allocated Time:	2.5 hrs
DAY 17: Sep 25, 2026 (Fri)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Advanced Linked List Operations
Theory / Pattern:	In-place group reversals, node swapping without extra memory
Target Problems:	Reverse Nodes in k-Group (#25), Copy List with Random Pointer (#138)
Allocated Time:	2.5 hrs
DAY 18: Sep 26, 2026 (Sat)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Stack Fundamentals & Matching
Theory / Pattern:	LIFO execution, balanced parenthesis state matching
Target Problems:	Valid Parentheses (#20), Min Stack (#155), Evaluate Reverse Polish Notation (#150)
Allocated Time:	2.5 hrs

DAY 19: Sep 27, 2026 (Sun)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Monotonic Stack - Next Greater Element
Theory / Pattern:	Maintaining monotonic decreasing/increasing order in stack
Target Problems:	Next Greater Element I (#496), Daily Temperatures (#739)
Allocated Time:	2.5 hrs
DAY 20: Sep 28, 2026 (Mon)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Weekend Focus: Monotonic Stack Applications
Theory / Pattern:	Histogram area calculations, trapped water mechanics
Target Problems:	Largest Rectangle in Histogram (#84), Trapping Rain Water (#42)
Allocated Time:	5.0 hrs
DAY 21: Sep 29, 2026 (Tue)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Weekend Revision & Timed Test 3
Theory / Pattern:	Weekly Review 3 + Timed Assessment on Linked Lists & Monotonic Stacks
Target Problems:	Test Set: 2 Mediums, 1 Hard
Allocated Time:	5.0 hrs
DAY 22: Sep 30, 2026 (Wed)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Binary Search - Standard Templates
Theory / Pattern:	Lower bound, upper bound, avoiding integer overflow
Target Problems:	Binary Search (#704), Search in Rotated Sorted Array (#33)
Allocated Time:	2.5 hrs
DAY 23: Oct 01, 2026 (Thu)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Binary Search on Answer Space
Theory / Pattern:	Monotonic predicate functions, searching boundary values
Target Problems:	Koko Eating Bananas (#875), Capacity To Ship Packages Within D Days (#1011)
Allocated Time:	2.5 hrs
DAY 24: Oct 02, 2026 (Fri)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Advanced Binary Search Patterns
Theory / Pattern:	Median of two sorted arrays, split array largest sum
Target Problems:	Find Minimum in Rotated Sorted Array (#153), Split Array Largest Sum (#410)
Allocated Time:	2.5 hrs

DAY 25: Oct 03, 2026 (Sat)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Tree Fundamentals & Traversals
Theory / Pattern:	Recursive DFS (Pre-order, In-order, Post-order), Iterative stack variants
Target Problems:	Binary Tree Inorder Traversal (#94), Maximum Depth of Binary Tree (#104)
Allocated Time:	2.5 hrs
DAY 26: Oct 04, 2026 (Sun)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Tree BFS & Level Order Traversal
Theory / Pattern:	Queue-based level-by-level evaluation, zigzag patterns
Target Problems:	Binary Tree Level Order Traversal (#102), Binary Tree Right Side View (#199)
Allocated Time:	2.5 hrs
DAY 27: Oct 05, 2026 (Mon)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Weekend Deep-Dive: Tree Properties
Theory / Pattern:	Lowest Common Ancestor, Tree construction from traversals
Target Problems:	Lowest Common Ancestor of a Binary Tree (#236), Construct Binary Tree from Preorder & Inorder (#105)
Allocated Time:	5.0 hrs
DAY 28: Oct 06, 2026 (Tue)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Weekend Revision & Timed Test 4
Theory / Pattern:	Weekly Review 4 + Diagnostic Mock Assessment 1
Target Problems:	Diagnostic Mock Interview Set: 2 Mediums, Verbal Explanation Practice
Allocated Time:	5.0 hrs
DAY 29: Oct 07, 2026 (Wed)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Binary Search Trees (BST)
Theory / Pattern:	BST properties, insertion, deletion, validation
Target Problems:	Validate Binary Search Tree (#98), Kth Smallest Element in a BST (#230)
Allocated Time:	2.5 hrs
DAY 30: Oct 08, 2026 (Thu)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Tree Path & Diameter Problems
Theory / Pattern:	Bottom-up recursion, subtree sum tracking
Target Problems:	Diameter of Binary Tree (#543), Binary Tree Maximum Path Sum (#124)
Allocated Time:	2.5 hrs

DAY 31: Oct 09, 2026 (Fri)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Heaps / Priority Queues Fundamentals
Theory / Pattern:	Binary heap representation, heapq in Python
Target Problems:	Kth Largest Element in an Array (#215), Top K Frequent Elements (#347)
Allocated Time:	2.5 hrs
DAY 32: Oct 10, 2026 (Sat)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Two Heaps Pattern
Theory / Pattern:	Maintaining streaming median using min-heap and max-heap
Target Problems:	Find Median from Data Stream (#295), Sliding Window Median (#480)
Allocated Time:	2.5 hrs
DAY 33: Oct 11, 2026 (Sun)	Phase 2: Core Linear & Non-Linear
Topic Focus:	K-Way Merge & Monotonic Queue
Theory / Pattern:	Merging sorted streams, sliding window maximums
Target Problems:	Merge k Sorted Lists (#23), Sliding Window Maximum (#239)
Allocated Time:	2.5 hrs
DAY 34: Oct 12, 2026 (Mon)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Weekend Mastery: Heaps & Complex Trees
Theory / Pattern:	Serialize and deserialize trees, hard heap problems
Target Problems:	Serialize and Deserialize Binary Tree (#297), Smallest Range Covering Elements from K Lists (#632)
Allocated Time:	5.0 hrs
DAY 35: Oct 13, 2026 (Tue)	Phase 2: Core Linear & Non-Linear
Topic Focus:	Weekend Revision & Timed Test 5
Theory / Pattern:	Weekly Review 5 + Timed Assessment on Trees & Heaps
Target Problems:	Test Set: 3 Mediums
Allocated Time:	5.0 hrs
DAY 36: Oct 14, 2026 (Wed)	Phase 3: Graphs & Recursion
Topic Focus:	Recursion & Backtracking Foundations
Theory / Pattern:	Decision trees, base cases, state backtracking
Target Problems:	Subsets (#78), Subsets II (#90)
Allocated Time:	2.5 hrs

DAY 37: Oct 15, 2026 (Thu)	Phase 3: Graphs & Recursion
Topic Focus:	Backtracking - Combinations & Permutations
Theory / Pattern:	Pruning search space, handling duplicate elements
Target Problems:	Permutations (#46), Combination Sum (#39)
Allocated Time:	2.5 hrs
DAY 38: Oct 16, 2026 (Fri)	Phase 3: Graphs & Recursion
Topic Focus:	Backtracking - Grid & Puzzle Search
Theory / Pattern:	2D matrix paths, constraint satisfaction
Target Problems:	Word Search (#79), N-Queens (#51)
Allocated Time:	2.5 hrs
DAY 39: Oct 17, 2026 (Sat)	Phase 3: Graphs & Recursion
Topic Focus:	Graph Representation & BFS
Theory / Pattern:	Adjacency lists, level-by-level shortest path in unweighted graphs
Target Problems:	Clone Graph (#133), Number of Islands (#200)
Allocated Time:	2.5 hrs
DAY 40: Oct 18, 2026 (Sun)	Phase 3: Graphs & Recursion
Topic Focus:	Graph DFS & Connected Components
Theory / Pattern:	Visited tracking, cycle detection in undirected graphs
Target Problems:	Max Area of Island (#695), Surrounding Regions (#130)
Allocated Time:	2.5 hrs
DAY 41: Oct 19, 2026 (Mon)	Phase 3: Graphs & Recursion
Topic Focus:	Weekend Deep-Dive: Advanced Graph Traversals
Theory / Pattern:	Bipartite graph checking, 0-1 BFS with Deque
Target Problems:	Is Graph Bipartite? (#785), Shortest Path in Binary Matrix (#1091)
Allocated Time:	5.0 hrs
DAY 42: Oct 20, 2026 (Tue)	Phase 3: Graphs & Recursion
Topic Focus:	Weekend Revision & Timed Test 6
Theory / Pattern:	Weekly Review 6 + Timed Assessment on Backtracking & Basic Graphs
Target Problems:	Test Set: 2 Mediums, 1 Hard
Allocated Time:	5.0 hrs

DAY 43: Oct 21, 2026 (Wed)	Phase 3: Graphs & Recursion
Topic Focus:	Cycle Detection & Topological Sort (Kahn's)
Theory / Pattern:	In-degree tracking, DAG ordering
Target Problems:	Course Schedule (#207), Course Schedule II (#210)
Allocated Time:	2.5 hrs
DAY 44: Oct 22, 2026 (Thu)	Phase 3: Graphs & Recursion
Topic Focus:	Topological Sort via DFS
Theory / Pattern:	Post-order DFS state coloring (Unvisited, Visiting, Visited)
Target Problems:	Alien Dictionary (#269), Find Eventual Safe States (#802)
Allocated Time:	2.5 hrs
DAY 45: Oct 23, 2026 (Fri)	Phase 3: Graphs & Recursion
Topic Focus:	Disjoint Set Union (DSU / Union-Find)
Theory / Pattern:	Path compression, union by rank
Target Problems:	Number of Connected Components in Undirected Graph (#323), Redundant Connection (#684)
Allocated Time:	2.5 hrs
DAY 46: Oct 24, 2026 (Sat)	Phase 3: Graphs & Recursion
Topic Focus:	DSU Advanced Applications
Theory / Pattern:	Dynamic connectivity, grid components
Target Problems:	Accounts Merge (#721), Number of Islands II (#305)
Allocated Time:	2.5 hrs
DAY 47: Oct 25, 2026 (Sun)	Phase 3: Graphs & Recursion
Topic Focus:	Shortest Path - Dijkstra's Algorithm
Theory / Pattern:	Weighted graphs with non-negative edge weights using heapq
Target Problems:	Network Delay Time (#743), Path with Minimum Effort (#1631)
Allocated Time:	2.5 hrs
DAY 48: Oct 26, 2026 (Mon)	Phase 3: Graphs & Recursion
Topic Focus:	Weekend Deep-Dive: Shortest Path Variations
Theory / Pattern:	Cheapest Flights Within K Stops, Bellman-Ford/Floyd-Warshall theory
Target Problems:	Cheapest Flights Within K Stops (#787), Swim in Rising Water (#778)
Allocated Time:	5.0 hrs

DAY 49: Oct 27, 2026 (Tue)	Phase 3: Graphs & Recursion
Topic Focus:	Weekend Revision & Timed Test 7
Theory / Pattern:	Weekly Review 7 + Mid-Program Diagnostic Mock Interview
Target Problems:	Mid-Program Mock: 2 Mediums, 1 System Design / DSA crossover
Allocated Time:	5.0 hrs
DAY 50: Oct 28, 2026 (Wed)	Phase 3: Graphs & Recursion
Topic Focus:	Minimum Spanning Tree (MST)
Theory / Pattern:	Prim's & Kruskal's Algorithms
Target Problems:	Min Cost to Connect All Points (#1584), Critical Connections in a Network (#1192)
Allocated Time:	2.5 hrs
DAY 51: Oct 29, 2026 (Thu)	Phase 3: Graphs & Recursion
Topic Focus:	Trie (Prefix Tree) Foundations
Theory / Pattern:	TrieNode structure, insert, search, startsWith
Target Problems:	Implement Trie (Prefix Tree) (#208), Design Add and Search Words Data Structure (#211)
Allocated Time:	2.5 hrs
DAY 52: Oct 30, 2026 (Fri)	Phase 3: Graphs & Recursion
Topic Focus:	Trie Applications & Hard Problems
Theory / Pattern:	Bitwise XOR tries, dynamic word searching
Target Problems:	Word Search II (#212), Maximum XOR of Two Numbers in an Array (#421)
Allocated Time:	2.5 hrs
DAY 53: Oct 31, 2026 (Sat)	Phase 3: Graphs & Recursion
Topic Focus:	Bit Manipulation - Basics & Bitmasking
Theory / Pattern:	XOR properties, bit shifts, clearing/setting bits
Target Problems:	Single Number (#136), Number of 1 Bits (#191), Counting Bits (#338)
Allocated Time:	2.5 hrs
DAY 54: Nov 01, 2026 (Sun)	Phase 3: Graphs & Recursion
Topic Focus:	Bitmasking Applications
Theory / Pattern:	Representing sets as integers in DP/Backtracking
Target Problems:	Subsets using Bitmasking, Minimum XOR Sum of Two Arrays (#1879)
Allocated Time:	2.5 hrs

DAY 55: Nov 02, 2026 (Mon)	Phase 3: Graphs & Recursion
Topic Focus:	Weekend Mastery: Advanced Graph & Bit Manipulation
Theory / Pattern:	Complex graph patterns, bitmask DP prep
Target Problems:	Reconstruct Itinerary (#332), Shortest Path Visiting All Nodes (#847)
Allocated Time:	5.0 hrs
DAY 56: Nov 03, 2026 (Tue)	Phase 3: Graphs & Recursion
Topic Focus:	Weekend Revision & Timed Test 8
Theory / Pattern:	Weekly Review 8 + Timed Assessment on Graphs, Tries & Bitmasking
Target Problems:	Test Set: 3 Mediums
Allocated Time:	5.0 hrs
DAY 57: Nov 04, 2026 (Wed)	Phase 4: DP & Greedy
Topic Focus:	Greedy Strategy Fundamentals
Theory / Pattern:	Local optimal choices leading to global optimum, proof techniques
Target Problems:	Assign Cookies (#455), Gas Station (#134)
Allocated Time:	2.5 hrs
DAY 58: Nov 05, 2026 (Thu)	Phase 4: DP & Greedy
Topic Focus:	Greedy Intervals - Overlap & Merging
Theory / Pattern:	Sorting by start/end times, interval consolidation
Target Problems:	Merge Intervals (#56), Non-overlapping Intervals (#435)
Allocated Time:	2.5 hrs
DAY 59: Nov 06, 2026 (Fri)	Phase 4: DP & Greedy
Topic Focus:	Greedy Intervals - Scheduling & Meeting Rooms
Theory / Pattern:	Min platforms/rooms allocation
Target Problems:	Meeting Rooms II (#253), Minimum Number of Arrows to Burst Balloons (#452)
Allocated Time:	2.5 hrs
DAY 60: Nov 07, 2026 (Sat)	Phase 4: DP & Greedy
Topic Focus:	Dynamic Programming - 1D Core
Theory / Pattern:	Memoization vs Tabulation, identifying subproblems
Target Problems:	Climbing Stairs (#70), House Robber (#198), House Robber II (#213)
Allocated Time:	2.5 hrs

DAY 61: Nov 08, 2026 (Sun)	Phase 4: DP & Greedy
Topic Focus:	1D DP - Subsequences & Partitioning
Theory / Pattern:	Unbounded vs bounded choices
Target Problems:	Coin Change (#322), Word Break (#139), Decode Ways (#91)
Allocated Time:	2.5 hrs
DAY 62: Nov 09, 2026 (Mon)	Phase 4: DP & Greedy
Topic Focus:	Weekend Focus: Longest Increasing Subsequence (LIS)
Theory / Pattern:	$O(N^2)$ DP to $O(N \log N)$ binary search solution
Target Problems:	Longest Increasing Subsequence (#300), Russian Doll Envelopes (#354)
Allocated Time:	5.0 hrs
DAY 63: Nov 10, 2026 (Tue)	Phase 4: DP & Greedy
Topic Focus:	Weekend Revision & Timed Test 9
Theory / Pattern:	Weekly Review 9 + Timed Assessment on Greedy & 1D DP
Target Problems:	Test Set: 3 Mediums
Allocated Time:	5.0 hrs
DAY 64: Nov 11, 2026 (Wed)	Phase 4: DP & Greedy
Topic Focus:	2D Grid DP - Paths & Minimum Cost
Theory / Pattern:	Grid traversals with directional moves
Target Problems:	Unique Paths (#62), Minimum Path Sum (#64)
Allocated Time:	2.5 hrs
DAY 65: Nov 12, 2026 (Thu)	Phase 4: DP & Greedy
Topic Focus:	2D Grid DP - Advanced Constraints
Theory / Pattern:	Dungeon Game, grid paths with obstacles
Target Problems:	Unique Paths II (#63), Dungeon Game (#174)
Allocated Time:	2.5 hrs
DAY 66: Nov 13, 2026 (Fri)	Phase 4: DP & Greedy
Topic Focus:	Knapsack DP - 0/1 Knapsack Pattern
Theory / Pattern:	Subset sum, equal partition matching
Target Problems:	Partition Equal Subset Sum (#416), Target Sum (#494)
Allocated Time:	2.5 hrs

DAY 67: Nov 14, 2026 (Sat)	Phase 4: DP & Greedy
Topic Focus:	Knapsack DP - Unbounded Knapsack
Theory / Pattern:	Infinite supply items, rod cutting
Target Problems:	Coin Change II (#518), Combination Sum IV (#377)
Allocated Time:	2.5 hrs
DAY 68: Nov 15, 2026 (Sun)	Phase 4: DP & Greedy
Topic Focus:	String DP - Longest Common Subsequence (LCS)
Theory / Pattern:	Two-pointer string DP matrix alignment
Target Problems:	Longest Common Subsequence (#1143), Edit Distance (#72)
Allocated Time:	2.5 hrs
DAY 69: Nov 16, 2026 (Mon)	Phase 4: DP & Greedy
Topic Focus:	Weekend Mastery: String DP & Palindromes
Theory / Pattern:	Palindromic subproblems, interval DP
Target Problems:	Longest Palindromic Substring (#5), Palindromic Substrings (#647), Distinct Subsequences (#115)
Allocated Time:	5.0 hrs
DAY 70: Nov 17, 2026 (Tue)	Phase 4: DP & Greedy
Topic Focus:	Weekend Revision & Timed Test 10
Theory / Pattern:	Weekly Review 10 + Advanced Diagnostic Mock Interview
Target Problems:	Advanced Mock: 1 Medium, 1 Hard with full interview communication
Allocated Time:	5.0 hrs
DAY 71: Nov 18, 2026 (Wed)	Phase 4: DP & Greedy
Topic Focus:	DP with Stock Trading State Machines
Theory / Pattern:	State transitions (Hold, Unhold, Cooldown, Fee)
Target Problems:	Best Time to Buy and Sell Stock with Cooldown (#309), Best Time to Buy and Sell Stock IV (#188)
Allocated Time:	2.5 hrs
DAY 72: Nov 19, 2026 (Thu)	Phase 4: DP & Greedy
Topic Focus:	Interval DP Pattern
Theory / Pattern:	Solving subproblems on sub-ranges [i...j]
Target Problems:	Burst Balloons (#312), Matrix Chain Multiplication
Allocated Time:	2.5 hrs

DAY 73: Nov 20, 2026 (Fri)	Phase 4: DP & Greedy
Topic Focus:	Tree DP Fundamentals
Theory / Pattern:	Subtree state computation, bottom-up DP on trees
Target Problems:	House Robber III (#337), Tree Diameter via DP
Allocated Time:	2.5 hrs
DAY 74: Nov 21, 2026 (Sat)	Phase 4: DP & Greedy
Topic Focus:	Bitmask DP
Theory / Pattern:	Traveling Salesperson Problem (TSP) style states
Target Problems:	Can I Win (#464), Matchsticks to Square (#473)
Allocated Time:	2.5 hrs
DAY 75: Nov 22, 2026 (Sun)	Phase 4: DP & Greedy
Topic Focus:	DP Optimization Techniques
Theory / Pattern:	Space optimization from $O(N^2)$ to $O(N)$ using rolling arrays
Target Problems:	Space-optimized LCS & Knapsack re-implementations
Allocated Time:	2.5 hrs
DAY 76: Nov 23, 2026 (Mon)	Phase 4: DP & Greedy
Topic Focus:	Weekend Focus: Complex DP Consolidation
Theory / Pattern:	Systematic approach to identifying DP transitions under pressure
Target Problems:	Regular Expression Matching (#10), Wildcard Matching (#44)
Allocated Time:	5.0 hrs
DAY 77: Nov 24, 2026 (Tue)	Phase 4: DP & Greedy
Topic Focus:	Weekend Revision & Timed Test 11
Theory / Pattern:	Weekly Review 11 + Timed Assessment on Advanced DP
Target Problems:	Test Set: 2 Mediums, 1 Hard
Allocated Time:	5.0 hrs
DAY 78: Nov 25, 2026 (Wed)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Mixed Array & Two Pointer Speed Run
Theory / Pattern:	Speed solving, verbalizing optimal trade-offs within 15 mins
Target Problems:	3 Unseen Medium Problems under 20-min timers
Allocated Time:	2.5 hrs

DAY 79: Nov 26, 2026 (Thu)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Mixed Graph & Tree Blitz
Theory / Pattern:	Identifying BFS vs DFS vs DSU instantly
Target Problems:	3 Unseen Medium/Hard Graph Problems
Allocated Time:	2.5 hrs
DAY 80: Nov 27, 2026 (Fri)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Mixed DP & Greedy Blitz
Theory / Pattern:	Distinguishing Greedy vs DP choices
Target Problems:	3 Unseen Medium DP Problems
Allocated Time:	2.5 hrs
DAY 81: Nov 28, 2026 (Sat)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Hard Problem Strategies
Theory / Pattern:	Partial credit strategies, simplifying constraints
Target Problems:	Sliding Window Maximum (#239), Alien Dictionary (#269)
Allocated Time:	2.5 hrs
DAY 82: Nov 29, 2026 (Sun)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Full Interview Simulation 1
Theory / Pattern:	Real-time interview simulation (45 mins per problem)
Target Problems:	Simulated Round 1: 2 Medium Problems + Verbal Communication
Allocated Time:	2.5 hrs
DAY 83: Nov 30, 2026 (Mon)	Phase 5: Final Bootcamp
Topic Focus:	Weekend Bootcamp: Full Mock Coding Marathon
Theory / Pattern:	4-hour continuous coding round simulation covering 4 mixed topics
Target Problems:	Full Mock Set: Array, Graph, DP, System-style DSA
Allocated Time:	5.0 hrs
DAY 84: Dec 01, 2026 (Tue)	Phase 5: Final Bootcamp
Topic Focus:	Weekend Bootcamp: Weak Spot Surgical Repair
Theory / Pattern:	Targeted repair based on mistake log history
Target Problems:	Re-solving 5 lowest-confidence problems from log
Allocated Time:	5.0 hrs

DAY 85: Dec 02, 2026 (Wed)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Full Interview Simulation 2
Theory / Pattern:	Real-time interview simulation (MNC Tier-1 standard)
Target Problems:	Simulated Round 2: 1 Medium, 1 Hard Problem
Allocated Time:	2.5 hrs
DAY 86: Dec 03, 2026 (Thu)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Edge Case & Bug Hunting Drills
Theory / Pattern:	Identifying off-by-one errors, null pointers, integer overflows
Target Problems:	10 buggy code snippets to audit and fix in Python
Allocated Time:	2.5 hrs
DAY 87: Dec 04, 2026 (Fri)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Python DSA Speed Optimization
Theory / Pattern:	In-place ops, list comprehensions vs generators, fast I/O
Target Problems:	Refactoring past code for maximum speed and clean syntax
Allocated Time:	2.5 hrs
DAY 88: Dec 05, 2026 (Sat)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Ultimate Pattern Cheat Sheet Memorization
Theory / Pattern:	Trigger word recognition (e.g. 'Shortest Path Unweighted' -> BFS)
Target Problems:	Pattern drill quiz across all 50 major patterns
Allocated Time:	2.5 hrs
DAY 89: Dec 06, 2026 (Sun)	Phase 5: Final Bootcamp
Topic Focus:	Bootcamp: Final Full-Length Mock Interview
Theory / Pattern:	External or simulated peer interview with full evaluation rubric
Target Problems:	2 Mixed Medium/Hard Problems with edge-case testing
Allocated Time:	5.0 hrs
DAY 90: Dec 07, 2026 (Mon)	Phase 5: Final Bootcamp
Topic Focus:	DAY 90: GRADUATION & MNC READINESS ASSESSMENT
Theory / Pattern:	Final Assessment, Complete Progress Audit, Next 30-Day Maintenance Plan
Target Problems:	Final Mixed Assessment Test + Review of Readiness Checklist
Allocated Time:	5.0 hrs

5. Pattern Recognition Cheat Sheet

Problem Clue / Trigger	Primary Pattern	Data Structure / Method
Contiguous array + Target sum / Max length	Sliding Window / Prefix Sum	Two Pointers / Hash Map
Sorted array + Pair target / Duplicate removal	Two Pointers	Opposite / Fast-Slow Pointers
Top K elements / Streaming values	Heap / Priority Queue	Min-Heap / Max-Heap (heapq)
Next greater/smaller element / Histogram area	Monotonic Stack	Stack storing indices
Shortest path on unweighted graph/grid	Breadth-First Search (BFS)	collections.deque
Shortest path on weighted graph (non-negative)	Dijkstra's Algorithm	Min-Heap + Distance Array
Dependencies / Task scheduling / DAG order	Topological Sort	Kahn's (In-degree) / DFS State
Dynamic connectivity / Disjoint groups	Union-Find (DSU)	Parent array + Rank/Size
Prefix matching / Autocomplete / XOR max	Trie	TrieNode with Dict children
Count ways / Min cost to reach state / Subarrays	Dynamic Programming	1D/2D DP array or Memoization

Idiomatic Python Template: Top-Down DP with LRU Cache

```
from functools import lru_cache
def solve_dp_problem(arr):
    @lru_cache(maxsize=None)
    def dp(idx, state_param):
        # Base Cases
        if idx >= len(arr):
            return 0
        # Transition Choices
        option1 = dp(idx + 1, state_param)
        option2 = arr[idx] + dp(idx + 1, new_state)
        return max(option1, option2)
    return dp(0, initial_state)
```

6. Mistake Log & Spaced Repetition System

Never resolve a problem without logging your failure mode. Categorize every error using these 5 categories:

- **1. Concept Gap:** Didn't know the underlying algorithm or data structure property.
- **2. Pattern Recognition Failure:** Knew the concept but failed to map problem clues to the pattern.
- **3. Logic Error:** Correct pattern, but incorrectly implemented condition/transition (e.g., off-by-one).
- **4. Edge Case Failure:** Solution failed on empty input, single element, negative numbers, or integer bounds.
- **5. Time/Space Management Failure:** Solved correctly but exceeded time limit (TLE) or memory limit (MLE).

7. MNC Interview Readiness Final Checklist

[]	Readiness Criterion	Target Standard
[]	Problem Solving Velocity	Easy: <10 mins Medium: 20-25 mins Hard: 40 mins
[]	Complexity Articulation	Instantly state Big-O Time & Space before typing code
[]	Python Built-in Fluency	Zero hesitation using heapq, collections, bisect

[]	Verbal Communication	Thinking out loud continuously; leading interviewer through solution tree
[]	Dry-Running Discipline	Testing code against edge cases manually before pressing Run/Submit
[]	Pattern Library Mastery	Can identify core pattern for 90%+ of unseen Medium problems in <3 mins